

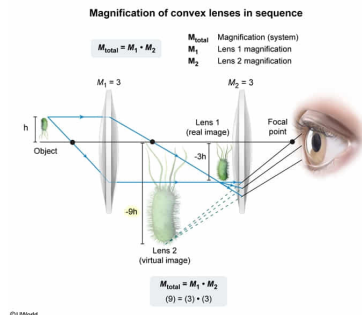
Two convex lenses are placed in series such that the image produced by first lens becomes the object of the second lens. If each lens is capable of producing a three-fold magnified image, what is the maximum magnification that this two-lens system can provide?

- ☐ A. 1.5 [2%]
☐ B. 3.0 [6%]
☐ C. 6.0 [17%]
☒ D. 9.0 [74%]

Correct

73% answered correctly

Explanation:



In optics, **magnification** refers to the ability of lenses to produce images that are smaller or larger than a real object that is placed at some distance in front of the lens. The expression for magnification (M) relates the object height (h_o) to the image height (h_i):

$$M = \frac{h_i}{h_o}$$

Magnification is not a fixed property of lenses. Both lens type and object placement influence the magnification of an image. For example, a **convex lens** may produce a **magnified real image** if the object is placed at a distance from the lens greater than lens focal length.

Image magnification may also occur within lens systems comprised of multiple lenses; this type of setup is common inside microscopes and telescopes. For multiple lenses **placed in series** (ie, not in direct contact), the magnification of the lens series is a **product** of each lens' individual magnification:

$$M_{total} = M_1 \cdot M_2$$

In this question, two convex lenses each with an individual magnification of 3 are placed in series and spaced such that the image of the object produced by the first lens becomes the image magnified by the second lens. Applying the equation for total magnification for this system gives:

$$M_{total} = M_1 \cdot M_2$$

Therefore, the total magnification produced by such a two-lens setup would be equal to 9.

(Choices A and B) Depending on lens characteristics and object placement, a two-lens system *may* yield a value for magnification that is less than or equal to the magnification of each lens. However, the question refers to *maximum* possible magnification.

(Choice C) Total magnification must be found by multiplying (*not* adding) the magnification value of the first lens by that of the second lens. Addition is used to find the total **lens strength S** of multiple lenses placed immediately adjacent to one another, where total power equals the sum of individual lens powers.

Educational objective:

Magnification is the phenomenon in which a lens produces an image that is enlarged or shrunken relative to the original object. Multi-lens systems arranged in series provide a combined magnification equal to the product of all individual lens magnifications.

Time spent: 0 sec
 Subject:
 Physics

QID: 402045
 System:
 Light and Sound